

“OPTICS OF EYE,VISUAL ACUITY AND ERRORS OF REFRACTION”

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“LEARNING OBJECTIVES”

- At the end of lecture ,students must be able to
- Describe the focusing power of the eye.
- Determine the power of lenses.
- Define accommodation.
- Describe Visual acuity
- Recognize nearsightedness and farsightedness and determine its correction

OPTICS OF THE EYE

THE EYE AS A CAMERA

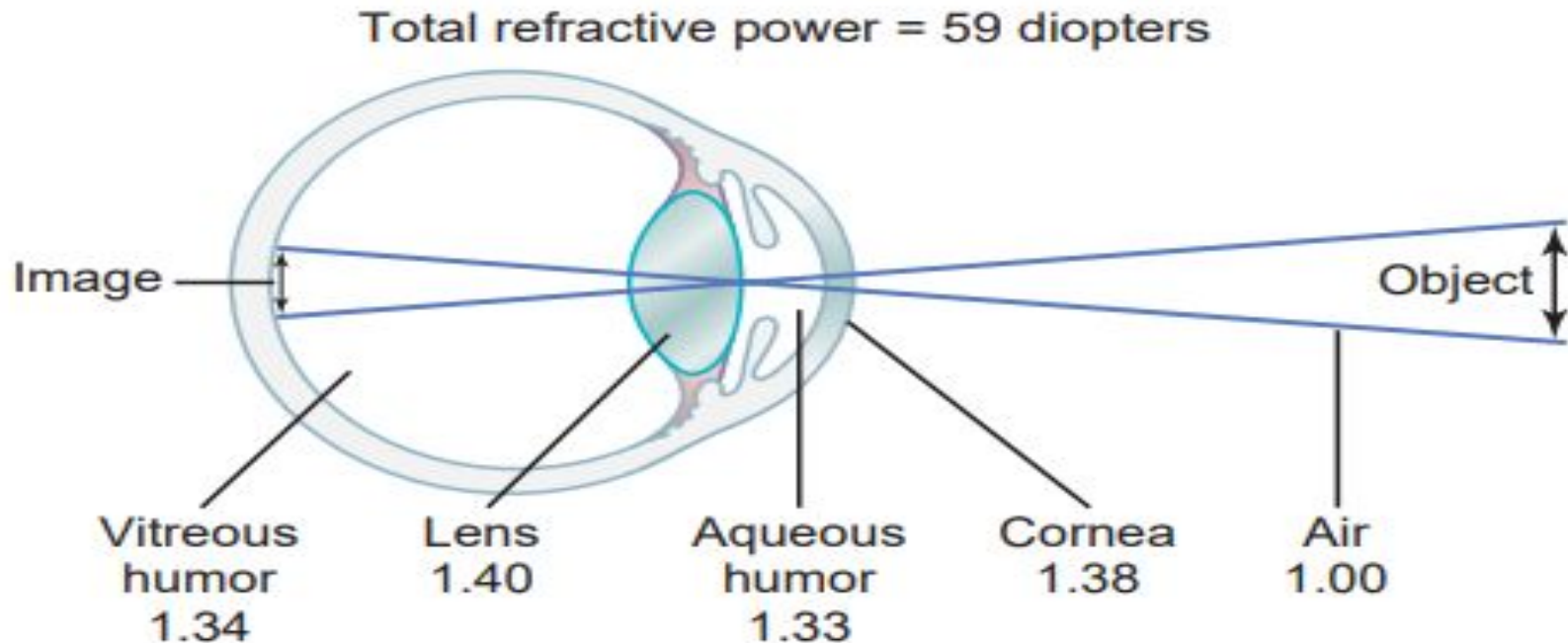


Figure 50-9. The eye as a camera. The numbers are the refractive indices.

Optics of eye

Reduced eye:

If all the refractive surfaces of the eye are algebraically added together & then considered to be one single lens, the optics of the normal eye may be simplified and represented schematically as a “reduced eye.” This is useful in simple calculations.

The total dioptic power of the eye is about +59 (+60) D at rest out of which about +44 D is contributed by cornea & +16 D by the crystalline lens.

Calculation of 59 diopter

- Focal point is 17 mm behind lens =0.017 meter
- $1/0.017=58.8$

Focusing power of the Eye

- Most of the refractive power of the eye results from the surface of the cornea.
- A diopter is a measure of the power of a lens.
- 1 diopter is the ability to focus parallel light rays at a distance of 1 meter, it is a measure of power of lenses.
- $\text{Diopter} = 1/\text{focal length in meters}$ i.e the power of a lens with focal length 0.5 meters is 2 (more convex).
- The retina is considered to be 17 mm behind the refractive center of the eye.
- Therefore, the eye has a total refractive power of 59 diopters ($1000/17$)

ACCOMODATION

- Accommodation is the adjustment of eye to see either near or distant objects clearly. It is the process by which light rays from near objects or distant objects are brought to a focus on sensitive part of retina. It is achieved by various adjustments made in the eyeball.

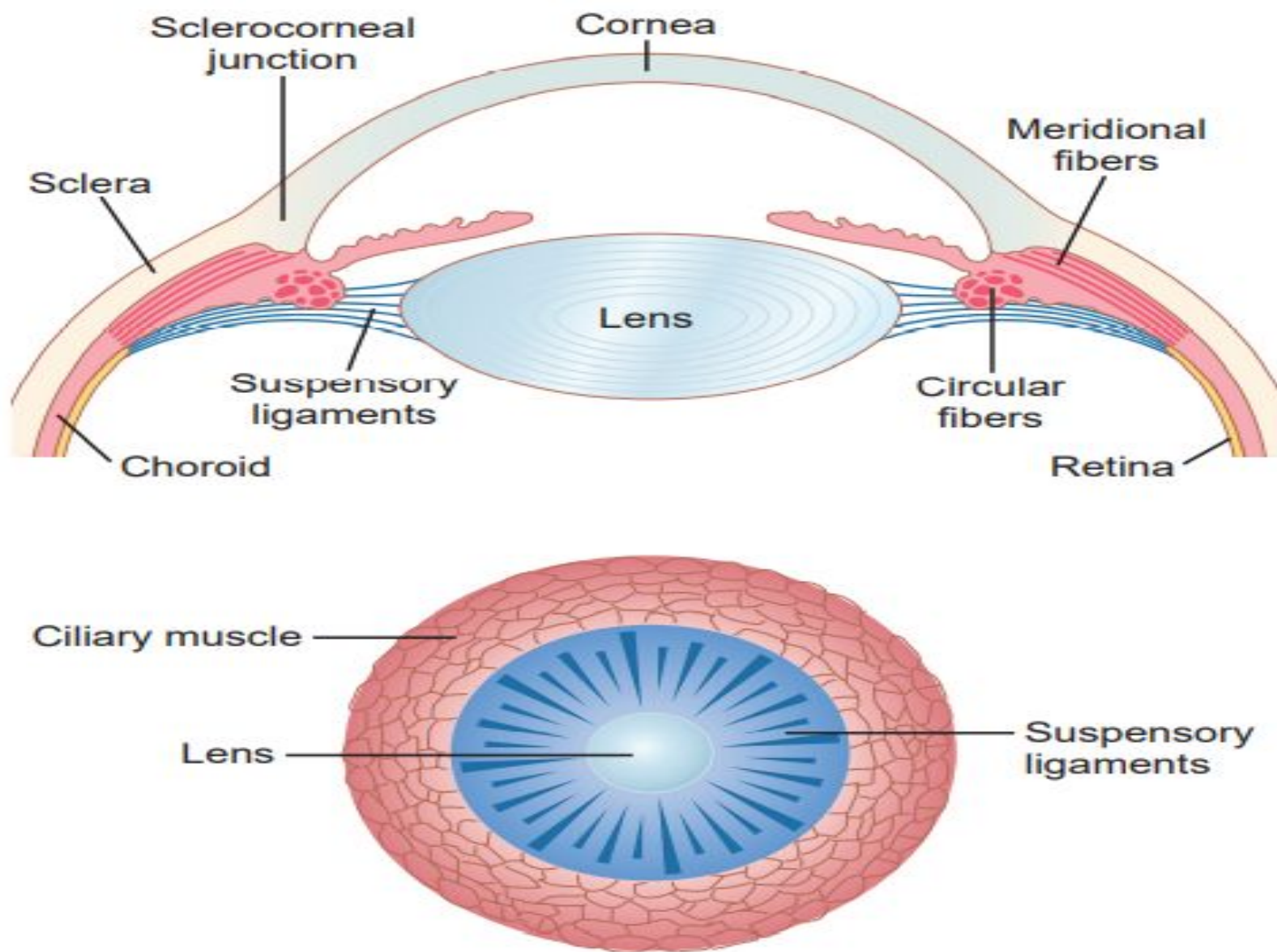


Figure 50-10. Mechanism of accommodation (focusing).

VISUAL ACUITY

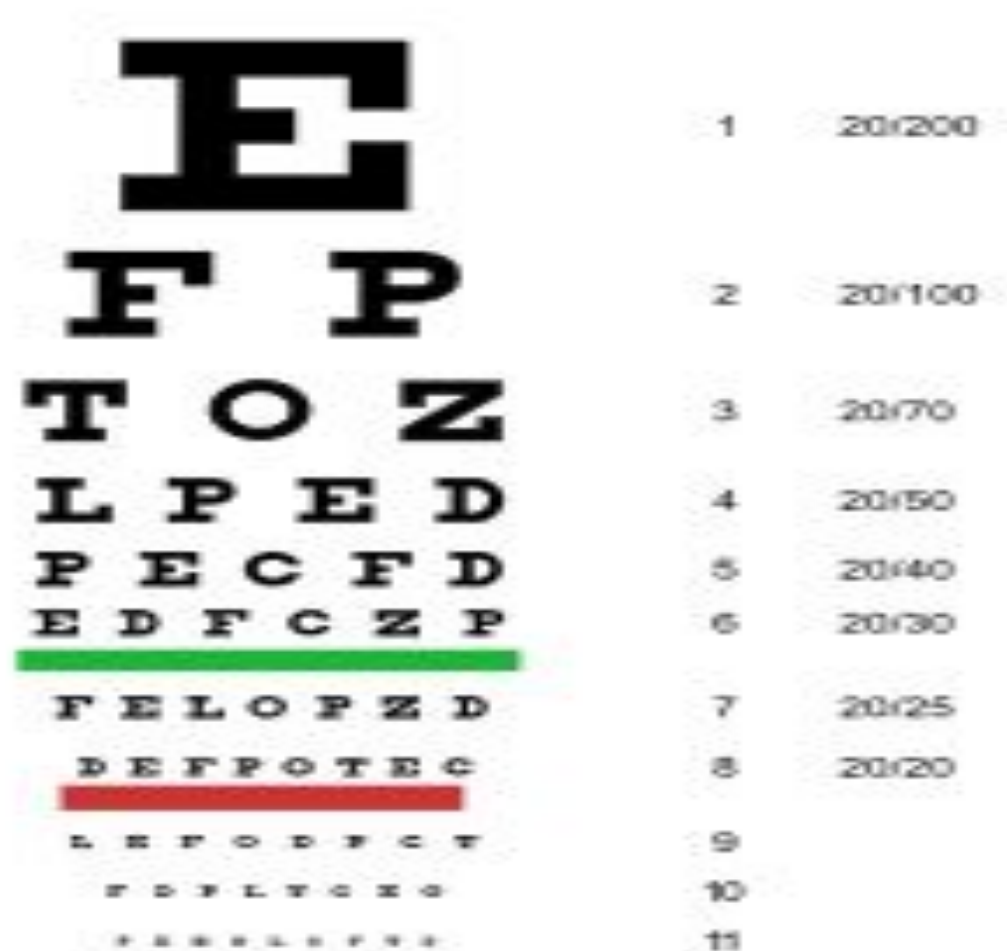
DEFINITION: The ability of eye to determine the precise shape and details of the object is defined as visual acuity. It is also defined as the ability to recognize the separateness of two objects placed together. Cones of retina are responsible for acuity of vision. Visual acuity is highly exhibited in fovea centralis, which contains only cones. It is greatly reduced during the refractory errors.

Visual acuity (VA)

- Acuteness or clearness of vision.
- dependent on the sharpness of the retinal focus within the eye.
- the sensitivity of the interpretative faculty of the brain.



Typical Snellen chart used for visual acuity testing.



“VISUAL ACUITY”

- To test visual acuity a **black symbols** on a **white background** are used (for maximum contrast)
- A sufficient distance allowed to approximate infinity in the way the lens attempts to focus.
- **Twenty feet** is essentially infinity from an optical perspective
- The difference in optical power required to focus at 20 feet versus infinity is only **0.164 diopters**.

“VISUAL ACUITY”

- **An eye exam lenses of varying powers are used to precisely correct for refractive errors.**
- **Letters are normally used (as in the classic Snellen chart) as most people will recognise them but other symbols (such as a letter E, facing in different directions) can be used instead.**

“VISUAL ACUITY”

- In the term "**20/20 vision**"
- The numerator refers to the distance in feet between the subject and the chart.
- The denominator is the distance at which the lines that make up those letters would be separated by a visual angle of 1 arc minute.
- For the lowest line that is read by an eye with no refractive error (or the errors corrected) is usually 20 feet.

“VISUAL ACUITY”

- The metric equivalent is **6/6 vision** where the distance is 6 meters.
- This means that at 20 feet or 6 meters.
- A typical human eye, able to separate 1 arc minute, can resolve lines with a spacing of about 1.75mm

“VISUAL ACUITY”

- 20/20 vision can be considered nominal performance for human distance vision.
- 20/40 vision can be considered half that acuity for distance vision.
- 20/10 vision would be twice normal acuity.

“VISUAL ACUITY”

- 20/x number does not directly relate to the eyeglass prescription required to correct vision.
- It does not specify the nature of the problem with the lens.
- Only the resulting performance.
- Instead an eye exam seeks to find the prescription that will provide at least 20/20 vision.

“VISUAL ACUITY”

- Visual acuity, or resolving power, is the property of cones.
- To resolve detail, the eye's optical system has to project a focused image on the fovea, a region inside the macula having the highest density of cone photoreceptor cells (the only kind of photoreceptors existing on the fovea), thus having the highest resolution and best color vision.
- Acuity and color vision, despite being mediated by the same cells, are different physiologic functions that do not interrelate except by position

“PHYSIOLOGY OF VISUAL ACUITY”

- The visual cortex is the part of the cerebral cortex in the posterior part of the brain responsible for processing visual stimuli, called the occipital lobe.
- The central 10° of field (approximately the extension of the macula) is represented by at least 60% of the visual cortex.
- Many of these neurons are believed to be involved directly in visual acuity processing.

“PHYSIOLOGY OF VISUAL ACUITY”

- The eye is connected to the visual cortex by the optic nerve coming out of the back of the eye.
- The two optic nerves come together behind the eyes at the optic chiasma where about half of the fibers from each eye cross over to the opposite side and join fibers from the other eye representing the corresponding visual field.
- The combined nerve fibers from both eyes forming the optic tract.

“PHYSIOLOGY OF VISUAL ACUITY”

- This ultimately forms the physiological basis of binocular vision.
- The tracts project to a relay station in the midbrain called the lateral geniculate nucleus which is part of the thalamus.
- Then to the visual cortex along a collection of nerve fibers called the optic radiations.

“PHYSIOLOGY OF VISUAL ACUITY”

- Visual acuity is affected by the size of the pupil.
- Optical aberrations of the eye that decrease visual acuity are at a maximum when the pupil is largest (about 8 mm) occurs in low-light conditions.
- When the pupil is small (1–2 mm), image sharpness may be limited by diffraction of light by the pupil (see diffraction limit).
- Between these extremes is the pupil diameter that is generally best for visual acuity in normal, healthy eyes; this tends to be around 3 or 4 mm

NORMAL VISION

- Visual acuity depends upon how accurately light is focused on the retina (mostly the macular region).
- The integrity of the eye's neural elements, and the interpretative faculty of the brain.
- Normal visual acuity is frequently considered to be what was defined by Snellen as the ability to recognize an optotype when it subtended 5 minutes of arc, that is Snellen's chart 20/20 feet, 6/6 meter, 1.00 decimal or 0.0 logMAR.

Refractive error

- A **refractive error**, or **refraction error**, is an error in the focusing of light by the eye and a frequent reason for reduced visual acuity.
- An eye that has no refractive error when viewing a distant object is said to have *emmetropia* or be *emmetropic*

Refractive error

- Refractive errors are frequently categorized as spherical errors and cylindrical errors:
- When the optics are too powerful for the length of the eyeball (this can arise from a cornea with too much curvature or an eyeball that is too long), one has myopia.
- When the optics are too weak for the length of the eyeball (this can arise from a cornea with not enough curvature or an eyeball that is too short), one has hyperopia.

ERRORS OF REFRACTION

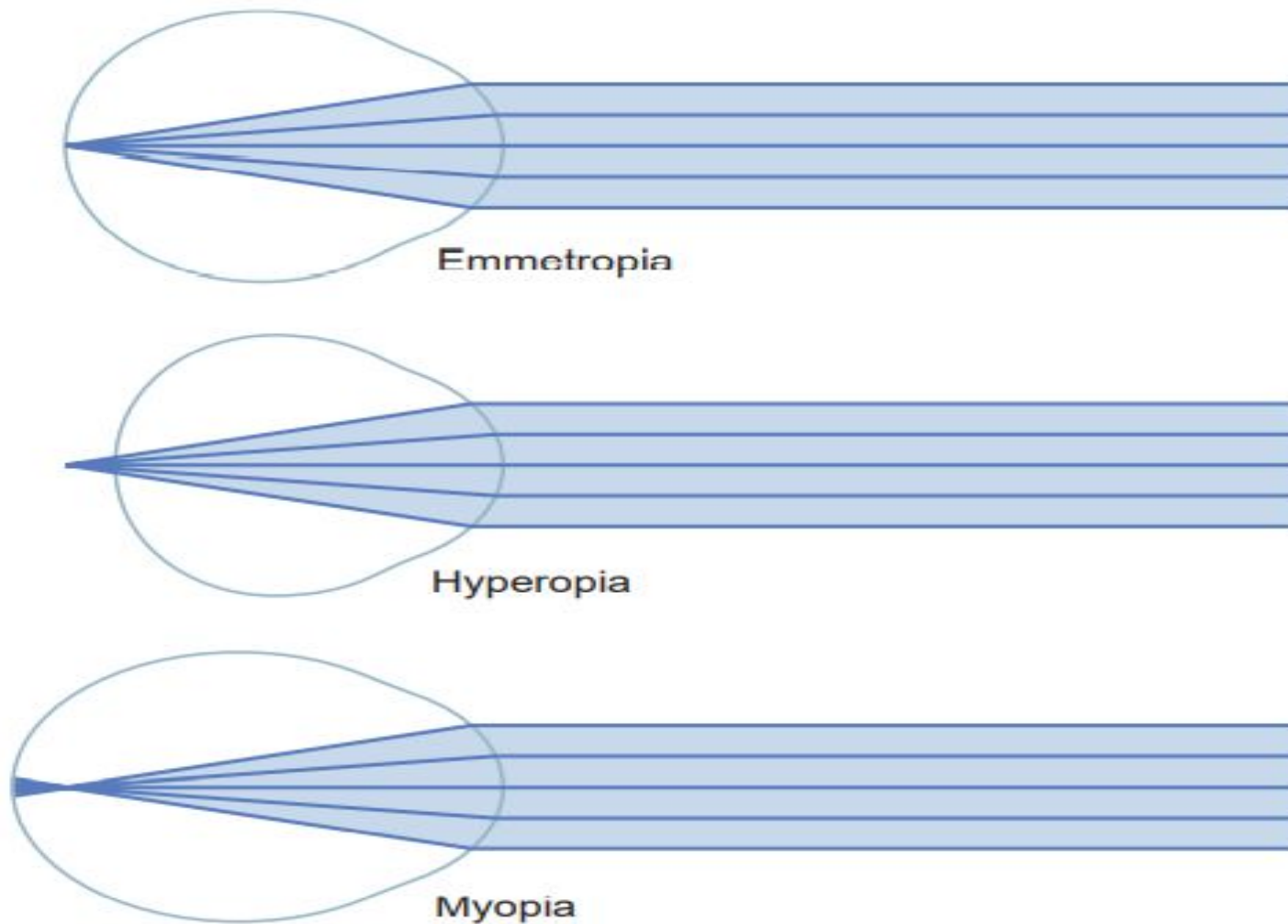


Figure 50-12. Parallel light rays focus on the retina in emmetropia, behind the retina in hyperopia, and in front of the retina in myopia.

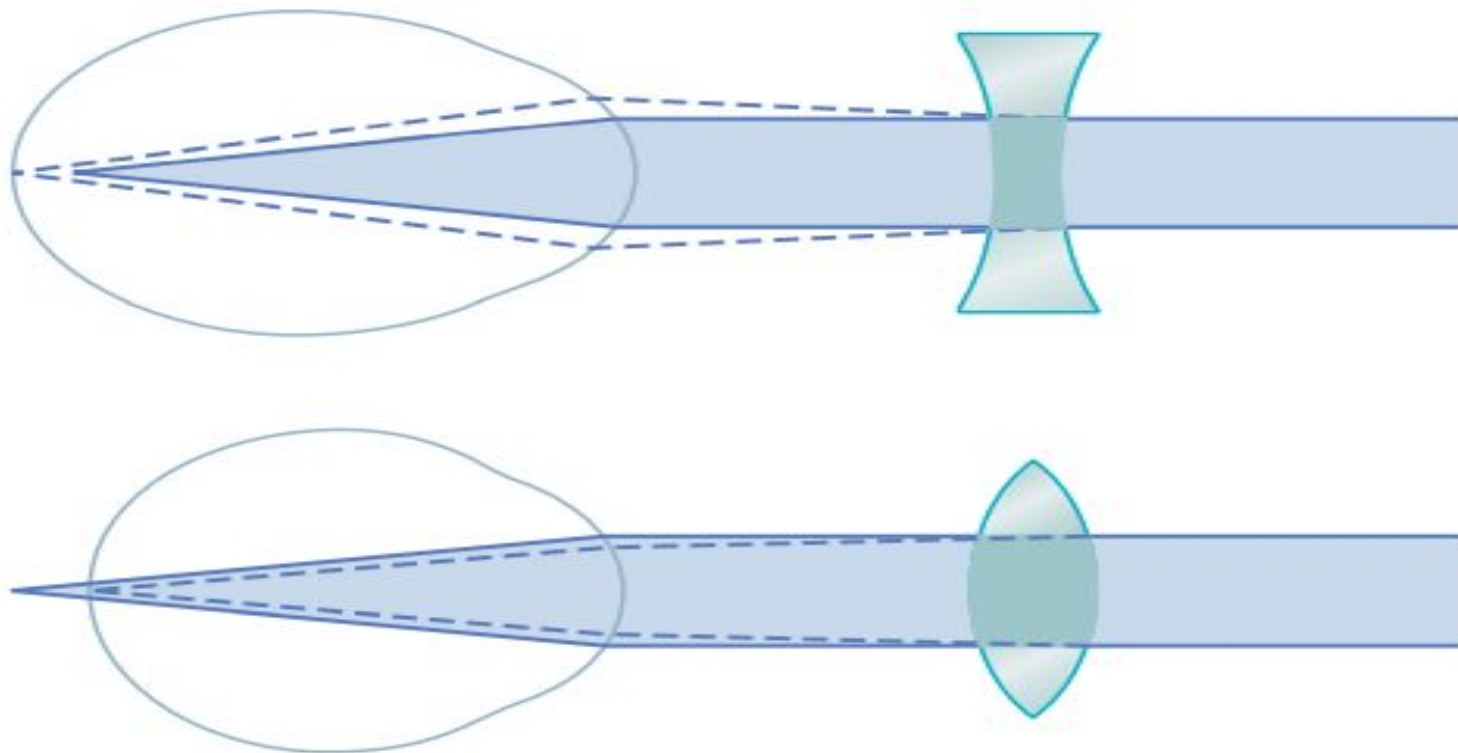


Figure 50-13. Correction of myopia with a concave lens (*top*) and correction of hyperopia with a convex lens (*bottom*).

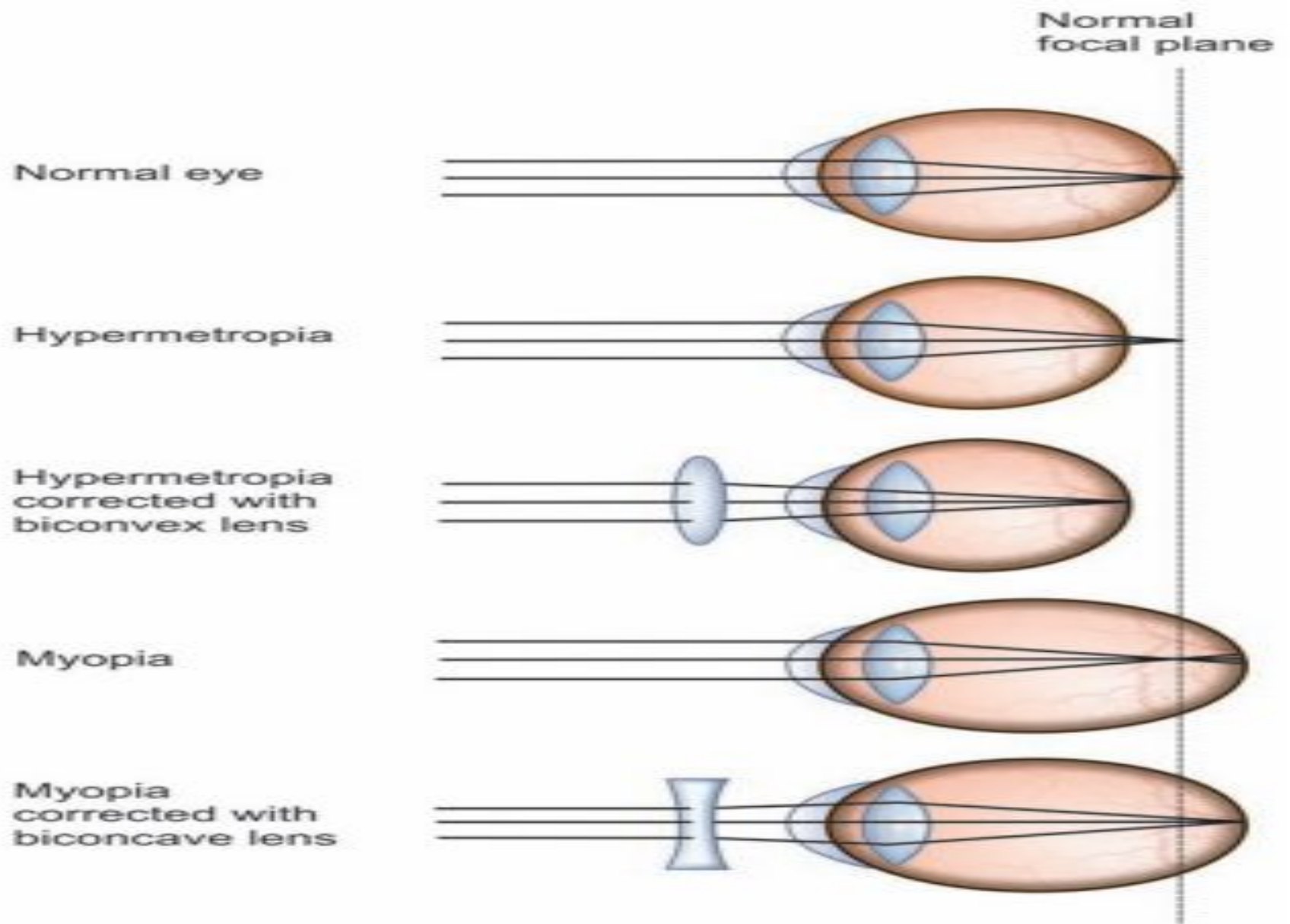


FIGURE 171.1: Errors of refraction

Refractive error

Spherical errors occur when the optical power of the eye is either too large or too small to focus light on the retina. People with refraction error frequently have blurry vision.

Causes: Refractive errors are thought to occur due to a combination of genetic and environmental factors. Trauma or ocular disorders such as keratoconus may induce refractive errors

Refractive error

- **Cylindrical errors** occur when the optical power of the eye is too powerful or too weak across one meridian.
- It is as if the overall lens tends towards a cylindrical shape along that meridian.
- The angle along which the cylinder is placed is known as the axis of the cylinder, while 90 degrees away from the axis is known as the meridian of the cylinder.
- People with a simple astigmatic refractive error see contours of a particular orientation as blurred, but see contours with orientations at right angles as clear.
- When one has a cylindrical error, one has astigmatism.

Management

For those who are symptomatic,

- Glasses,
- Contact lenses,
- Refractive surgery,
- Or a combination of the three are typically used.